Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/15/2022
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	06/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2023 per NSSC (Ed., 5/16/22) NOTE: End date changed to 6/30/2022 per F. Hernandez/ARC (Ed., 8/19/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/2020) NOTE: Extended to 7/31/2020 per NSSC information (Ed., 5/15/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro.] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will be launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
	The Seedling Growth-2 experiment was performed in the European Modular Cultivation System (EMCS) on board the International Space Station (ISS). The original experiment setup included two runs, each 6 days long. The experiment resulted in good germination rates for Run 1 and Run 2, and the results were sufficient to a give good science return. In addition to seed germination, the Principal Investigator (PI) teams noted robust growth and phototropic curvature in all runs. Seedling Growth-3 (SG3) was the first experiment to employ the European Space Agency (ESA)-built FixBoxes, and the operations by the astronauts with this new hardware were nominal. Following the landing of SpaceX5, SG3 cassettes with frozen seedlings were recovered and processed at NASA Ames Research Center (ARC) by the PI team and ARC staff. At the end of this processing, frozen seedlings were removed from the cassettes and were placed into microfuge tubes. These specimens were shipped to Madrid for RNA extraction and gene profiling studies by Co-PI F.J. Medina and his group. In addition, in some cases, seedlings were removed from the FixBoxes and shipped to Spain for further processing. After identification of differentially expressed genes by RNAseq methods, a Generally Applicable Gene-Set (GAGE)-Pathway Analysis was conducted between microgravity and 1g conditions to determine if any molecular pathways were found to be significantly enriched with genes (identified as differentially expressed) to give a more holistic view of the effects of reduced gravity on plant molecular pathways. GAGE analysis revealed that 8 pathways were found to be significantly enriched (2 upregulated, 6 downregulated; FDR < 0.1) with genes identified as differentially expressed in the microgravity vs 1g group. These pathways included photosynthesis and photosynthetic antenna proteins as being significantly downregulated. We also had some interesting results on reduced or partial gravity from our space experiments on the ISS. Seedlings grew for six days under different gravity levels, namely microgravity, several intermediate partial-g levels, and 1g, and were subjected to irradiation with blue light for the last 48 hours. RNA was extracted from 20 samples for subsequent RNAseq analysis. Transcriptomic analysis was performed using the HISAT2-Stringtie-DESeq pipeline. Differentially expressed genes were further characterized for global responses using GEDI tool, gene networks, and for Gene Ontology (GO) enrichment. Principal Component Analysis (PCA) of the replicates: The clustering of the replicates is consistent. There is a very clear separation among the three g-levels in the samples exposed to red light photostimulation (µgrl, Marsrl and grrrl) but it is not well distinguished between the Marsd and grrd samples. To elaborate on this, we have applied two different approaches to perform the comparisons in the transcriptomics data. First, to investigate the effect of different gravity levels on A. thaliana seedlings, we have compared the transcriptomes of samples grown in microgravity or Mars gravity level to the same light condition in 1g GRR transcriptome as the reference; µgd-grrd, Marsd-grrd, µgrl-grrrl; Marsrl-grrrl. Next, to dissect the effect of red light at different g levels, we have compared the transcriptomes of seedlings grown in the same g levels but in different light conditions; µgrl-µgd; Marsrl-Marsd, grrrl-grrd. In both cases, we have used a q-value < 0.05 and a threshold fold change of Log2 FC +/- 1.5. In the gravity level comparisons, more than 600 differentially expressed genes (DEGs) were upregulated in µgd, µgrl, Marsrl whereas in Marsd only half of this number (372) were upregulated. These results show that transcriptomes of the seedlings photostimulated by red light presented a similar number of upregulated DEGs at both gravity levels, but in seedlings grown in the dark, twice as many DEGs were upregulated in microgravity than in Mars gravity. Approximately half of the upregulated DEGs are common for each gravity level independently of being exposed to darkness or photostimulation (253 and 175 in microgravity and Mars gravity, respectively). The low number of upregulated genes in the Marsd sample is in accordance with a small size of meristematic nucleoli in this condition, suggesting a reduced rate of protein biosynthesis, meaning that the transcriptome status is reflected at the proteome

Task Progress:	level. This effect could be a result of the upregulation of Ovate Family Protein 10 (OFP10), a transcription repressor upregulated only in the Marsd condition (Log2FC 2; all Log2FC values given in the text are statistically significant; q-value < 0.05). In the Gene Ontology (GO) analysis, we used the total number of upregulated or downregulated DEGs to look for common and specific altered molecular functions in each condition. We observed that, in the upregulated genes, there was a clear clustering of common categories by g-levels rather than by light conditions. Among the categories upregulated in all conditions, response to osmotic stress, wounding, and cellular response to hypoxia could be identified. Photosynthesis categories are highly upregulated in microgravity, but not at Mars g-level. It is surprising that this category is equally upregulated in both light conditions – the red-light photostimulated sample and the seedlings grown in darkness for two days. Another strongly upregulated category in microgravity was oxidative phosphorylation. On the other hand, categories specific for Mars conditions (both light treatments) include response to ethylene, drug, defense response, and positive regulation of biosynthetic processes and organ growth. The ethylene pathway, together with Ethylene Response Factors (ERFs), are involved in a wide range of stress responses. Although it has been suggested previously that the detection of an ethylene response in spaceflight experiments was an effect of the ethylene accumulated from the previous experiments, any ethylene accumulation during this experiment would have been erased from the hardware by a flushing procedure performed before the initiation of the experiment, as well as by constant air flow by a connected gas removal module. The dysregulation of different hormonal routes in our samples suggests that, in microgravity, the seedlings do not address the external stress and take the option of growing by activating growth and proliferation-promoting auxin and cytokinin routes. On the other hand, in the Mars g samples, the growth-promoting routes are inhibited and abscisic acid (ABA), ethylene, and salicylic acid routes, known for their crucial role in response to osmotic, drought, and biotic stresses, are activated. These studies show the utility of spaceflight experiments in making discoveries in the fundamental biology of plants, which will be needed to fully utilize plants in bioregenerative life support systems in space missions. Since blue-light phototropism is robust in both roots and shoots, we studied expression of our five candidate genes with reverse transcription-polymerase chain reaction (RT-PCR) during this tropism in light-grown wild type (WT) seedlings. For three of the five genes (MTS, NFL, and ABC), there were changes in gene expression during the time course studies of blue-light phototropism. For MTS, the time course of curvature studies suggests that this gene attenuates phototropism in the shoots, and RT-PCR studies show a corresponding decrease in expression of this gene. MTS has been shown to be involved in organ growth and regulation of cell cycle, but this is the first report of it being involved in tropism pathways. Like the PIN-FORMED (PIN) family, ABC proteins have been shown to play a role in gravitropism and phototropism. ATP-BINDING CASSETTE (ABC) transporters are one of the largest superfamily of transmembrane transporter proteins that utilize adenosine triphosphate (ATP) to carry out translocation of substrates across membranes. In plants, ABC proteins shuttle a variety of substrates such as lipids, carboxylates, heavy metals, and hormones such as auxin, across a variety of biological membranes. Our time course of curvature studies suggest that ABC attenuates blue-light phototropism in roots. However, in contrast, RT-PCR experiments suggest that ABC gene expression increases during the same time. In some ways, the most important finding of our studies is that NFL plays a role in gravitropism and phototropism. NFL proteins are particularly fascinating because of their role as intermediate filaments (IF). While the existence of IFs in plants has been debated, new molecular and cytological evidence confirms that IFs do occur in plant cells. The importance of NFL also is suggested by data in the Klepikova Atlas (used in the TAIR database; https://), which provides tissue-specific profiling for many genes throughout the Arabidopsis thaliana genome. Thus, tissue-specific profiling indicates that NFL shows high expression in the roots of 1-day old seedlings, and moderate expression in aerial tissue. Many studies have shown that microtubules and the actin cytoskeleton play a major role in gravitropism and phototropism, but our space studies raise the possibility that IFs also are involved in these pathways.
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